

KVM & virsh Cheat Sheet

Create, start, stop and inspect KVM virtual machines (libvirt calls them "domains") from the shell.

Using it

```
sudo virsh COMMAND web1 ← most commands take the VM's name
export LIBVIRT_DEFAULT_URI=qemu:///system ← or this; else you see qemu:///session (looks empty)
```

Reading virsh list --all

```
$ sudo virsh list --all
 Id   Name      State
-----
 1    web1      running
 4    db1       paused
 -    test1     shut off
```

You see	It means
Id	Only while running (changes each start); - when off
running	Up and using CPU and RAM
paused	Frozen in RAM (suspend); resume continues it
shut off	Not running; uses disk space only

Everyday virsh commands

virsh ...	Does	virsh ...	Does
start web1	Boots it	suspend web1	Pauses it (resume continues)
shutdown web1	Asks the guest to shut down	autostart web1	Start at host boot (--disable)
reboot web1	Asks the guest to reboot	console web1	Serial console; exit: Ctrl-]
destroy web1	Pulls the plug (deletes nothing)	reset web1	Presses the reset button
dominfo web1	State, vCPUs, memory	edit web1	Edit XML (applies at next cold start)
domifaddr web1	IP addresses (see note below)	dumpxml web1	Prints the full XML config
domiflist web1	NICs: bridge, model, MAC	define web1.xml	Register a VM from XML
dombllklist web1	Disk files it uses	undefine web1	Forget the VM; disks kept

Bridged VM shows no IP? Add **--source agent** (needs **qemu-guest-agent** in the VM) or **--source arp**.

Attaching a VM to the bridge main

```
<interface type='bridge'>
  <source bridge='main' /> ← the host bridge
  <model type='virtio' /> ← fastest NIC type
</interface>
```

Add with **virsh edit web1** (a MAC is filled in), or live:

```
virsh attach-interface web1 bridge main
--model virtio --live --config
```

The VM joins your LAN and gets an IP from its DHCP.

Creating a VM with virt-install

```
sudo virt-install --name web1 --memory 4096 --vcpus 2 --disk size=40 \ ← new 40 GB qcow2 disk
  --cdrom /var/lib/libvirt/images/debian-12.iso --os-variant debian12 \
  --network bridge=main,model=virtio --graphics vnc --noautoconsole
```

Snapshots

virsh ...	Does	virsh ...	Does
snapshot-create-as web1 snap1	Take a snapshot	snapshot-revert web1 snap1	Go back to it
snapshot-list web1	List snapshots	snapshot-delete web1 snap1	Delete it

Disks, storage and networks

Command	Does
qemu-img info web1.qcow2	Format; virtual size (VM sees) vs disk size (really used)
qemu-img create -f qcow2 data.qcow2 40G	New thin 40 GB disk (grows as it fills)
qemu-img resize web1.qcow2 +20G	+20 GB, VM off (live: virsh blockresize)
virsh pool-list --all / vol-list default	Pools / their disks (/var/lib/libvirt/images)
virsh net-list --all	libvirt networks (default = NAT on virbr0)

Remember: **destroy** only pulls the plug; **undefine --remove-all-storage** is what deletes disks. After **qemu-img resize**, grow the partition inside the guest too.